

FormalA Development Roadmap

Phase 0 - Environment Setup

- Create project structure: /backend, /frontend, /exports, etc.
- Create and activate Python virtual environment
- Create requirements.txt and install dependencies (FastAPI, Transformers, etc.)
- Create FastAPI base app in main.py
- Run and test FastAPI at <http://127.0.0.1:8000>
- Test OpenRouter API connectivity using Mistral model (via OpenAI-compatible client)
- Test Hugging Face Transformers (e.g., summarization pipeline)
- Update roadmap to reflect use of OpenRouter instead of OpenAI

Phase 1 - Backend MVP

- Create CourseRequest and CourseResponse models (Pydantic)
- Create generator.py to handle AI logic using OpenRouter (Mistral)
- Create /generate-course endpoint in course.py using FastAPI
- Test /generate-course via Swagger UI (<http://127.0.0.1:8000/docs>)
- Return raw course content from AI (outline, summary, title)
- Plan to parse and structure the output into CourseResponse (later)

Phase 2 - Frontend MVP (Streamlit)

- Create Streamlit frontend for user to enter topic and level
- Connect frontend to FastAPI endpoint using requests or httpx
- Display course title, outline, and summary in clean UI
- Allow exporting results as Word/PDF (later)

Phase 3 - Enhancements and Testing

- Add QCM generation endpoint (using Hugging Face or prompt engineering)
- Improve summary generation with different model
- Add error handling and timeouts to API calls
- Add loading states and retry logic in frontend

Phase 4 - Advanced Features

- Implement CNFCPP sheet generator
- Generate PowerPoint output (python-pptx)
- Add practical exercises per module (AI generated)
- Integrate with DALL·E or image generation for visual assets

Phase 5 - Finalization & Deployment

- Write documentation (README.md + user PDF guide)
- Deploy app to cloud (Render, Railway, etc.)
- Polish UI and UX
- Present or record demo for final delivery